

Trekking App Project Report

1. Student Details

Name: Palak Verma

Roll No.: 23f1000489

Email: 23f1000489@ds.study.iitm.ac.in

2. Project Details

Project Title: Trekking Management System

Problem Statement:

To develop a web-based trekking management system that allows users to browse trekking packages, register, book treks, while enabling administrators to manage treks, staff, users, and bookings efficiently.

Approach:

The application is developed using Flask following an MVC-like structure. SQLite is used for data storage through SQLAlchemy. The system provides separate interfaces for administrators and users, supports authentication, trek booking, staff management, and booking history. Bootstrap, HTML, CSS and Jinja2 are used for the frontend.

3. AI/LLM Declaration

ChatGPT (GPT-5) was used for documentation assistance. The application logic, testing, and debugging were completed manually.

4. Technology and Framework Used

Technology	Purpose
Flask	Backend Framework
SQLAlchemy	Database ORM
Sqlite	Database
Jinja2	Template Engine
Html, CSS	FrontEnd
BootStrap	Responsive UI

5. Database Schema and ER Diagram

Main Tables:

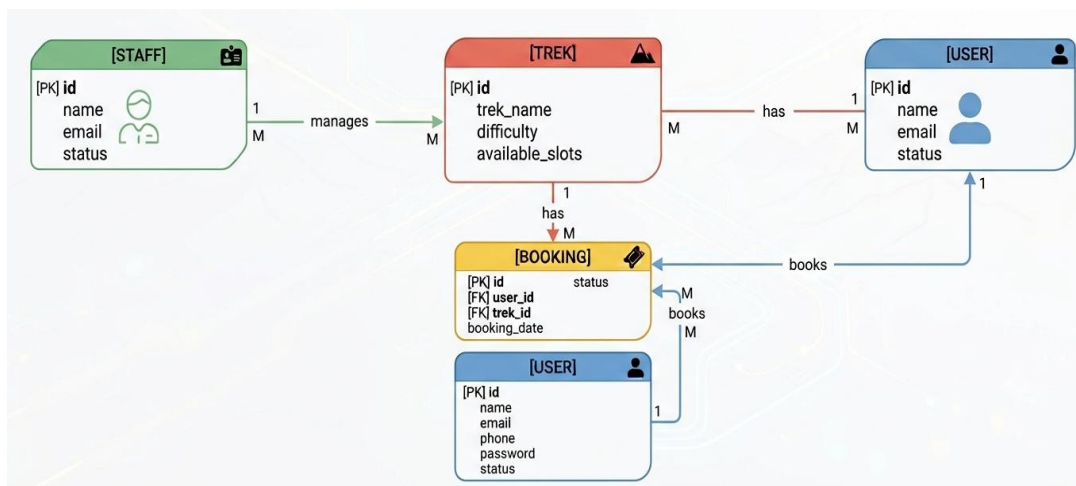
- User
- Staff
- Trek
- Booking

Relationships:

User → Booking (1:M)

Staff → Trek (1:M)

Trek → Booking (1:M)



6. API Routes(Major Routes)

Route	Method	Purpose
/login	GET/POST	Login
/register	GET/POST	Registration
/admin_dashboard	GET	Admin Dashboard
/user_dashboard/<id>	GET	User Dashboard
/staff_dashboard/<id>	GET	Staff Dashboard
/book_trek	POST	Book trek

7. Architecture and Features

Architecture:

- app.py – Entry point
- application/controllers.py – Route handling
- application/models.py – Database models
- templates/ – HTML templates

- static/ – CSS and images

Features:

- User registration and login
- Trek booking
- Admin dashboard
- Trek management
- Staff management
- Booking history
- Search functionality
- Responsive interface

8. Video Presentation

Link: <https://drive.google.com/file/d/1gZD8jDTMsLILXp9E1vY9S0ZAY-Jy7LzT/view?usp=sharing>